Verbal Ability and Theory of Mind in Internationally-Adopted Children: A Randomized Control Trial

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Background

- Internationally adopted (IA) children experience early social deprivation in orphanages or baby homes overseas
- IA children are at risk for negative social and cognitive developmental outcomes (Lawler & Gunnar, 2012)
- IA children experience significant delays in Theory of Mind (Tarullo, Bruce & Gunnar, 2007)
- ToM is the ability to attribute mental states to oneself and others
- ToM associated with:
  - Social competence
  - Quality of peer relationships (Capage & Watson, 2001)
  - Academic achievement (Lecce, Caputi, & Hughes, 2011)
  - Language ability (de Villiers, 2007)
  - Executive function (Carlson, Mandell, & Williams, 2004)
- Few studies have explored how ToM may be trained or improved

Aims

- Can ToM be improved by cognitive training interventions designed to improve core EF skills?
- Does verbal ability at baseline contribute to improvements in ToM?

The Current Study

- Randomized controlled trial (RCT) used to compare the effects of
  - executive function training (EF)
  - mindfulness-based training (MT)
  - a non-intervention control

  on ToM in IA children ages 6-10
- MT and EF interventions improve:
  - Selective attention
  - Cognitive flexibility
  - Impulse control (Diamond & Lee, 2011; Tang et al., 2012)
- Hypothesis: MT and EF have potential to improve ToM

Training Groups

Pre-testing Tasks

- Standard ToM task: Sally and Ann (Wimmer & Perner, 1983)
- Advanced ToM reasoning task (adapted from Happé, 1994)
- NIH Toolbox Picture vocab task (NU/NIH, 2012), age-adjusted

MT Intervention



EF Intervention



Post-testing Tasks

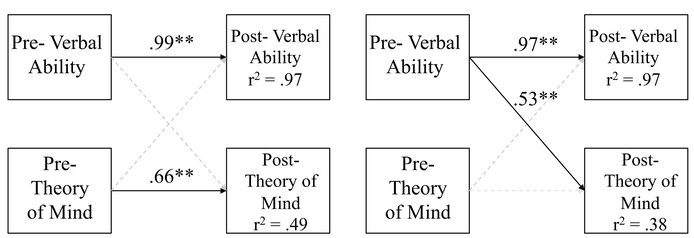
- Standard ToM task: Sally and Ann (Wimmer & Perner, 1983)
- Advanced ToM reasoning task (adapted from Happé, 1994)
- NIH Toolbox Picture vocab task (NU/NIH, 2012), age-adjusted

- Intervention groups: 12 one-hour training sessions over 6 weeks
- Home practice activities
- Interventions designed to improve:
  - inhibitory control
  - cognitive flexibility
  - selective attention
- Composite ToM score created by summing total number correct across both ToM tasks

Wave 1 Sample and Results

- **Structural Equation Modeling (Schumacker & Lomax, 1996) used to examine aims**

Figure 1. Wave 1 Final Models Examining Bidirectional Associations Between Children's Verbal Ability[a] and Theory of Mind: Presented Separately by Treatment Condition



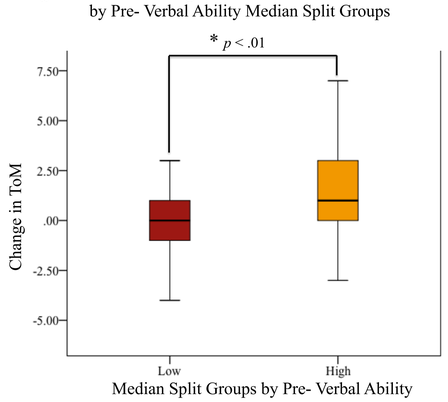
Note: χ^2 = (2, N = 66) = 1.29, p = .52, CFI = 1.00, TLI = 1.02, RMSEA = .00. *p < .05, **p < .01. All parameter estimates presented are standardized.

- **Wave 1 (Summer 2013)** total n=66: EF (n=21), MT (n=23), Control (n=22)
- **For intervention:** Higher pre-verbal ability associated with greater gains in ToM (p < .01)
- **For control:** no evidence for bidirectional association

Change in ToM by median split on pre-verbal ability:

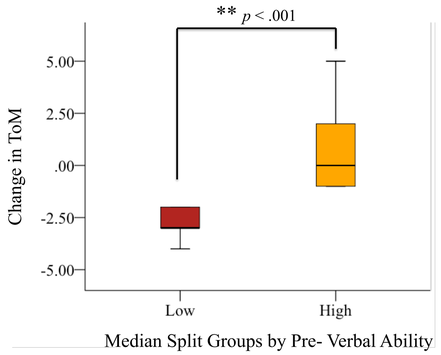
- **For combined intervention group:** children in the top 50% demonstrated a significant gain in ToM F(1, 41)=7.57, p < .01
- **For control group:** no significant difference in ToM from pre to post assessment F(1, 21)=.459, p = .51

Figure 2. Wave 1 Change in ToM within Treatment Condition by Pre- Verbal Ability Median Split Groups



Wave 2 Sample and Results

Figure 3. Wave 2 Change in ToM by Pre- Verbal Ability Median Split Groups



- **Wave 2** (Summer 2014) n=33:
  - EF (n=11)
  - MT (n=13)
  - control (n=9)
- Sample too small to examine aims with SEM

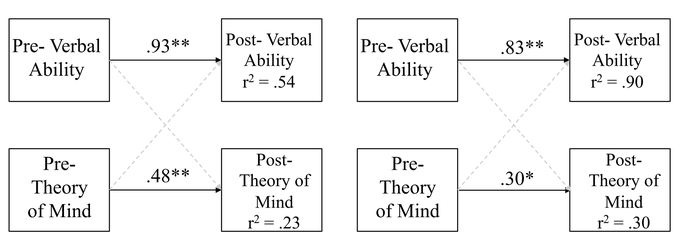
Change in ToM by median split on pre-verbal ability:

- Children in top 50% demonstrated significant gain in ToM F(1, 28) = 29.32, p <.001 compared to children in bottom 50%

Complete Sample Results

- **Complete sample** n=106: EF (n=35), MT (n=38), control (n=33)

Figure 4. Final Models Examining Bidirectional Associations in Complete Sample Between Children's Verbal Ability[a] and Theory of Mind: Presented Separately by Treatment Condition



Note: χ^2 = (2, N = 106) = 2.08, p = .35, CFI = 1.00, TLI = 1.00, RMSEA = .03. *p < .05, **p < .01. All parameter estimates presented are standardized.

- **Wave 1 findings not replicated**

Table 3
Significant Differences Between Wave 1 and Wave 2 (N = 106)

Variable	Mean (SD)	F	*p*-value
Pre-test Theory of Mind		1.65	.20
Wave 1	4.88 (2.11)		
Wave 2	5.42 (1.84)		
Post-test Theory of Mind		2.41	.12
Wave 1	5.55 (2.10)		
Wave 2	4.90 (1.40)		
Change in Theory of Mind from Pre to Post		4.52*	*.04*
Wave 1	.55 (2.15)		
Wave 2	-.47 (2.25)		
Pre-test Verbal Ability[b]		19.54**	*<.001*
Wave 1	92.47 (20.56)		
Wave 2	109.20 (10.36)		
Post-test Verbal Ability[b]		8.45*	*.005*
Wave 1	97.55 (23.56)		
Wave 2	110.60 (10.21)		
Pre-Adoption Time in Institutional Care (months)		8.63*	*.004*
Wave 1	12.57 (11.19)		
Wave 2	6.06 (8.91)		
Pre-Adoption Time in Foster Care (months)		1.24	.27
Wave 1	2.56 (4.42)		
Wave 2	3.53 (3.26)		
Pre-Adoption Time with Relatives (months)		46.77**	*<.001*
Wave 1	0 (0)		
Wave 2	.68 (.79)		
Pre-Adoption Time Unaccounted For (months)		4.41	*.04*
Wave 1	.28 (.71)		
Wave 2	1.32 (3.95)		
Age at Pre-test (months)		.09	.77
Wave 1	100.56 (19.89)		
Wave 2	99.42 (14.71)		
Sex (1=*male*, 2=*female*)		10.43*	*.002*
Wave 1	1.68 (.47)		
Wave 2	1.36 (.49)		

- Comparison of means (F-test) showed significant differences between waves on pre-adoption history and other study variables
- **Wave 2 characterized by:**
  - Higher baseline verbal ability
  - Less time spent in institutional care prior to adoption
  - More time spent with relatives prior to adoption
  - More time unaccounted for prior to adoption
  - More boys

Conclusions

- Mindfulness and EF trainings may help improve ToM skills among children exposed to early adversity
- Stronger baseline verbal abilities were associated with greater gains in ToM



Where will Sally look for her ball?

- Significant differences in preadoption history and other variables seemed to play a role in the inability to replicate results from Wave 1
- Verbal abilities are necessary but not sufficient for improving ToM
- Social-cognitive trainings may consider incorporating a vocabulary component to boost effects
- Contrary to previous research, a group training design may be effective in teaching ToM

Future Directions

- Use SMART intervention design
- Explore whether delivering curriculum one-on-one is more effective for children with lower verbal ability (Kloo & Perner, 2008)

Acknowledgements

This research was supported by the NIH/ NICHD Interdisciplinary Training Program in Cognitive Science (T32 HD007151) as well as small grants from the Institute of Child Development and the Center for Neurobehavioral Development, University of Minnesota.
We thank the families who participated in this research project.